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The Cost of Chronic Stress: Impacts of a Nonhabituating Stress Response on Metabolic Variables and Swimming Performance in Sturgeon

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ABSTRACT

Metabolic scope for activity (MSA) and critical swimming velocity (U_{crit}) were measured in green sturgeon exposed to two stressors daily for 28 consecutive days. The results were compared with unstressed fish in an effort to measure the “cost” of chronic stress. Chronic stress was simulated by exposing fish to a randomized order of acute stressors: a 5-min chasing stressor, a 10-min water depth reduction stressor, or a 5-min confinement stressor. The acute cortisol response to each stressor was initially determined, and the maintenance of that response was verified in 7-d intervals during the chronic stress regime. Exposure to the chronic stress regime resulted in a 25% reduction of MSA caused by significantly increased maintenance metabolic rate (0.27 ± 0.01 vs. 0.19 ± 0.02 mg O₂ h⁻¹ g⁻¹, chronic and control fish, respectively) but did not affect the U_{crit} of sturgeon. In addition, a 50% reduction in liver glycogen levels and a twofold increase of resting plasma glucose levels were measured in chronically stressed fish. We conclude that our chronic stress regime resulted in a significant maintenance cost to green sturgeon, possibly because of their inability to habituate to the stressors, but did not decrease their swimming performance.

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Introduction

The physiological and behavioral mechanisms that have evolved in vertebrates to cope with stressors, both perceived and real, have been the focus of many biologists since the introduction of the general adaptation syndrome by Selye (1950). Examination of the stress response in fish has revealed a complex set of coordinated responses (Idler and Truscott 1972; Wendelaar Bonga 1997; Barton et al. 2002; Greenberg et al. 2002) that have been highly conserved among fishes and other vertebrates (Greenberg et al. 2002). Catecholamines (epinephrine/norepinephrine) and the products of the hypothalamo-pituitary-interrenal (HPI) axis (corticotropin-releasing hormone, adrenocorticotropin [ACTH], and cortisol) are widely accepted as primary components and, therefore, indicators of an active stress response in fishes (Idler and Truscott 1972; Mazeaud et al. 1977; Donaldson 1981; Sumpter 1997; Wendelaar Bonga 1997), although additional regulatory compounds appear to be involved in the stress response of fishes (for review, see Barton et al. 2002). These diverse regulatory elements appear to enhance the effectiveness of the acute stress response, but their effects during times of prolonged stress are not as well studied.

Most studies that examine stress describe the acute response, which is known to involve immediate release of stored catecholamines followed by the synthesis and release of cortisol (Mazeaud et al. 1977; Gamperl et al. 1994; Wendelaar Bonga 1997). The initial perception of a stressor stimulates the release of catecholamines, which rapidly increase blood glucose concentrations through glycogenolysis (Larsson 1973; Randall and Perry 1992; Reid et al. 1998). Cortisol then stimulates catabolic processes to liberate amino acids (AA) and free fatty acids for the prolonged production of glucose through gluconeogenesis (Vijayan et al. 1994a, 1997; Milligan 1997). Cortisol also directly regulates gluconeogenesis by augmenting the transcription and activity of phosphoenolpyruvate carboxykinase, which is the rate-limiting enzyme in fish gluconeogenesis (Sathiyaa and Vijayan 2003; Vijayan et al. 2003). The majority of the free glucose synthesized by hepatic gluconeogenesis is sequestered and stored as hepatic and muscle glycogen, thereby repaying the debt incurred during the initial stress response (glycogenolysis; Vijayan et al. 1994a; Mommsen et al. 1999). However, cortisol may also enhance the breakdown of hepatic glycogen, a glucocorticoid-induced response that has been demonstrated in vitro (Vijayan et al. 1994b). These effects of cortisol seem

counterproductive because of the simultaneous upregulation of both gluconeogenesis and glycogenolytic mechanisms. It is suggested by Vijayan et al. (1994b) that the glucose liberated from the enhanced breakdown of glycogen is not used as a systemic glucose source and instead is used primarily to enhance hepatic metabolism. This cortisol-dependent scenario is thought to provide the animal with enhanced hepatic metabolic capabilities during times of prolonged stress. However, the effects of cortisol on liver glycogen content still are not clear (Barton and Schreck 1987; Vijayan et al. 1997; Mommsen et al. 1999). Other known secondary/tertiary components of the stress response include reduced or altered immune function (Maule et al. 1988; Dhabhar and McEwan 1997), reduced growth rate (Heath et al. 1993; Carr 2002), and reduced reproductive function (Carragher and Sumpter 1990; Campbell et al. 1994; Haddy and Pankhurst 1999; Lethimonier et al. 2002). The attenuation of these processes during acute stress is believed to allow reallocation of the liberated metabolic resources toward vital functions necessary for survival (Moberg 1985). Although these responses provide the animal with increased energetic resources to cope with a stressor, there is a short-term biological cost associated with the shifting of these resources. Barton and Schreck (1987) examined the metabolic cost associated with acute stress in juvenile steelhead (*Oncorhynchus mykiss irideus*) and concluded that the mechanisms involved in the acute stress response reduced metabolic scope for activity (MSA) by ca. 25%. Despite the association of the acute stress response with negative effects on the animal, there is little doubt regarding the adaptive significance of this neuroendocrine-systemic response on short-term survivability (Greenberg et al. 2002).

However, situations in which the stressors are persistent or chronic are commonly regarded as maladaptive (Selye 1950; Moberg 1985; Reid and Perry 1991; Barton et al. 2002). We know little about the organismal effects of chronic stress in fish, which is exacerbated by the variance in time periods that are considered chronic between studies. However, the available literature, which focuses on salmonid species, suggests an extension of these effects for acute stressors. Over time, these effects would reduce the fitness of the individual. Campbell et al. (1994) reported that two episodes of a 2-wk confinement stress reduced egg size and progeny survival in brown (*Salmo trutta*) and rainbow trout (*O. mykiss*). Similarly, food consumption, growth rate, and food conversion efficiency are reduced in rainbow trout receiving stresslike levels of exogenous cortisol (Gregory and Wood 1999). Clearly, chronic stress affects fitness on multiple levels, but if one considers the glucose metabolism paradigm, chronic stress would present a bioenergetic conundrum for the animal. During a period of chronic stress, the animal must compromise glycogen reserves, increase the catabolism of tissue protein (free AA), or potentially do both to upregulate repetitively the energetic resources required to maintain the components of the acute stress response. This increased use of energetic reserves is theorized to be a con-

tributing factor in the development of the prepathological state (Moberg 1985), which leaves the animal vulnerable to the development of pathology. Thus, if the animal became habituated during chronic exposure to stress, the energetic problem would be alleviated but the adaptive stress response would be compromised.

There is growing evidence suggesting that chronic stress alters the components of the acute stress response in salmonids. Barton et al. (1987) detected an inhibition of the HPI axis in juvenile rainbow trout fed cortisol-supplemented feed or physically stressed once daily for 10 wk. Furthermore, Shrimpton and Randall (1994) reported that the concentration of glucocorticoid receptors (GR) in the gill tissue of coho salmon (*Oncorhynchus kisutch*) were reduced during 18 d of exogenous cortisol treatment or endogenous cortisol increases through exposure to a single stressor per day. The reduction of gill GR resulted in reduced sensitivity of the gills to cortisol. These results suggest that chronic stress significantly alters the components of the acute stress response, but because of different experimental protocols and the limited number of species that have been examined, the implications are not fully understood.

In the following experiments, the effects of chronic stress were investigated in juvenile green sturgeon (*Acipenser medirostris*), a species of special concern in California, by exposing fish to two stressors per day for 28 consecutive days. We selected a 28-d time period to represent chronic stress in this study because previous experiments on salmonids report significant changes in fish exposed to cortisol for 7–14 d (Campbell et al. 1994; Shrimpton and Randall 1994; Perry et al. 1996; Gregory and Wood 1999). Because of the conserved nature of the stress response in vertebrates, we felt justified in using this period to adequately compare our results with the previously mentioned studies. Our purpose was to measure, for the first time, the responses of this phylogenetically ancient order of fishes to chronic stress and to compare the results with those of more derived teleosts in hopes of elucidating factors contributing to the decline of this species. We measured acute plasma cortisol responses to the individual stressors selected for the chronic stress regime; changes in plasma glucose, lactate, and cortisol in response to a single acute stressor at 7-d intervals during the chronic stress regime; and changes in maintenance metabolic rate (MMR), MSA, critical swimming velocity (U_{crit}), and liver glycogen concentrations after the completion of the 28-d chronic stress regime. We hypothesized that if green sturgeon failed to habituate to chronic stress, this would give rise to increased maintenance requirements, resulting in (1) increased MMR, (2) decreased MSA, (3) decreased U_{crit} , and (4) decreased liver glycogen.

Material and Methods

Fish Source and Care

Juvenile green sturgeon were obtained from artificially spawned, wild-caught brood stock from the Klamath River (Van

Eenennaam et al. 2001) and maintained in outdoor flow-through tanks supplied with 19°C well water under a natural photoperiod at the Center for Aquatic Biology and Aquaculture, University of California, Davis (38.6°N, 121.7°W). Fish were fed Silver Cup extruded, nonfloating trout chow at the rate of 1.0% body weight d^{-1} through a 24-h belt feeder. All fish were held in similar round (3.66-m-diameter) tanks until they were transferred into experimental tanks where they were allowed to acclimate for 2 wk before initiating experiments. These experiments used naive fish from two different year classes (2000 and 2001); however, year classes were not mixed within a single experiment. In all experiments, feeders were emptied 24 h before sampling to ensure all fish were in a postabsorptive metabolic state.

Experimental Procedures

Chronic Stress Regime. Chronic stress was simulated by subjecting fish ($n = 3$ per tank) to a randomized sequence of stressors (one of three) twice daily (1000 and 1600 hours) for 28 consecutive days. This chronic stress regime was adapted from the regime developed by Shrimpton and Randall (1994). The stressors included a 5-min chasing stressor that involved a small aluminum dip net with only incidental direct contact with the fish; a 10-min water depth reduction stressor that consisted of a rapid reduction in the water depth until the dorsal fins of the fish were slightly exposed to air followed by a gradual return to the original water level; or a 5-min confinement stressor, where the fish were held underwater in a large, fine-mesh tube net measuring at least 1.5 times the length of the fish. The weights of the fish were recorded at the beginning and end of the 28-d period, and fish received feed throughout the entire treatment period.

Acute Response to Chronic Stressors. The acute cortisol responses to each of the three stressors used in the chronic stress regime ($n = 6\text{--}7$ fish per stressor) were measured in age 0+ fish (year 2000 class; 1.0–1.4 kg live weight) acclimated to individual 1-m-diameter, indoor flow-through tanks. Laboratory lights were adjusted to simulate a natural photoperiod, and 24-h disk feeders provided food at a rate of 1.0% body weight d^{-1} . All other aspects of fish care were similar to the care of fish in the holding tanks. In-dwelling catheters were surgically implanted into the caudal vasculature 48 h before sampling, using the procedure described by Crocker and Cech (1998). Briefly, fish were rapidly anesthetized in a buffered MS 222 bath (350 mg L^{-1} , 9 g L^{-1} NaCl, and 0.42 g L^{-1} NaHCO_3) and transferred to a surgery table where the gills were continuously irrigated with a similar bath containing 150 mg L^{-1} MS 222. Using a modified 17-gauge Touhy needle as a guide, polyethylene tubing (Intramedic Clay-Adams PE-50, Becton Dickson, Sparks, MD) was inserted into the caudal vasculature and sutured in place. Cannulae were flushed with heparinized saline (50 units heparin mL^{-1}) and

tied off after sampling so that the fish could swim freely after returning to the original tank. The entire cannulation procedure was completed within 15 min. On the sampling day, blood was collected through the cannulae before exposure to the stressor (time 0) and 15, 30, 60, 120, and 180 min poststressor. The blood samples were stored on ice until centrifugation, and plasma was collected and stored at -80°C until analysis for cortisol.

Chronic Stress Habituation. Habituation of the acute stress response by exposure to the chronic stress regime (see “Chronic Stress Regime”) for 21 d was investigated by acclimating groups ($n = 3$) of age 1+ fish (2000 year class; 1.6–2.9 kg live weight) in round (2.1-m-diameter), outdoor flow-through tanks ($n = 5$) maintained as described for the holding tanks. Fish were exposed to the chronic stress regime daily except for the days (0, 7, 14, and 21) that their acute response to a water depth reduction stressor was determined. To determine the acute response, tanks were randomly assigned a “poststressor” sampling time of 0, 15, 30, 60, or 120 min poststressor the morning of each sampling day. Fish in the time-0 tank were always sampled first to avoid indirectly disturbing the tank, thereby assuring the measurement of accurate resting cortisol levels. After collection of the time-0 tank, the four remaining tanks were rapidly drained until the water level approached the fish’s dorsal surface but did not expose the fish to air. The water level remained depressed for 10 min and then was allowed to return to the initial level. Individual timers were set for each tank, and blood was sampled at the selected time poststressor according to the following protocol. All fish ($n = 3$) in the assigned tank were quickly netted and transferred into a buffered MS 222 anesthetic bath (350 mg L^{-1} , 6 g L^{-1} NaCl, and 0.42 g L^{-1} NaHCO_3) until undulations weakened and ventilatory movements ceased (ca. 1 min). Fish were then transferred to one of three V-shaped tables for blood sampling by way of cardiac puncture with heparinized syringes and 20-gauge needles. All blood samples were collected within 5 min of initial tank disturbance before plasma cortisol increases caused by the sampling techniques were detectable (Wedemeyer et al. 1990; Gamperl et al. 1994). Blood was processed as previously described.

Determination of MMR, AMR, MSA, and U_{crit} . Groups ($n = 3$) of age 0+ green sturgeon (year 2001 class; 0.7–1.6 kg) were chronically stressed in 1-m-diameter, indoor flow-through tanks ($n = 5$) maintained as described in “Acute Responses to Chronic Stressors.” Two of the three fish in each tank were used for the measurement of MSA and U_{crit} , and one was used for liver glycogen measurements. The initiation of each tank of fish into the chronic stress regime was staggered by 3 d to facilitate a continuous availability of fish for the postchronic stress regime experiments and to assure that all fish would be tested within 2 d of completing the chronic stress regime. Con-

trol fish were held at similar densities in a 2-m-diameter, outdoor flow-through tank maintained as described for the holding tanks.

After the completion of the stress regime, fish were placed into the Brett-type swim-tunnel respirometer (ca. 655 L, of plastic and stainless steel construction with a variable speed motor and propeller) no later than 1600 hours and allowed to acclimate at a velocity of 10 cm s^{-1} until 0800–0900 hours the next morning to provide adequate time for the fish to recover from the transfer stress and adjust to the swimming chamber. The acclimation speed (10 cm s^{-1}), at which green sturgeon would remain calm in the apparatus and not need to swim actively to maintain position, was determined by pilot experiments. To assure that the water inside of the respirometer remained equilibrated with the aerated water bath, two large-capacity, submersible water pumps forced water into the respirometer through an opening equipped with a ball valve. The opposite side of the loop had an identical opening to allow water to return to the bath.

The morning of the experiment the respirometer oxygen partial pressure (Po_2) was adjusted, if needed, to 140–160 mmHg by bubbling oxygen into the water bath. Closing the ball valves and switching off the flushing pumps sealed the respirometer, and a water sample was collected for initial Po_2 (Cameron Instruments BGM 200) measurement through microbore Tygon tubing, stopcock, and glass syringe. Water samples (in duplicate) were removed every 30 min until the Po_2 decreased by at least 10 mmHg, and ventilatory rates and tail-beat frequencies were monitored (remote video camera and monitor). If resting ventilatory rates and/or tail-beat frequencies were above 80 or 10 beats min^{-1} , respectively, fish were not used in the experiment. The MMR (oxygen consumption) was calculated by converting Po_2 into oxygen content (CO_2) with a nomogram (Green and Carritt 1967) and using the equation of Cech (1990) for closed respirometers. Bacterial respiration was not accounted for because weekly 60-min “blank” runs resulted in no Po_2 decreases, and the entire respirometer was bleached weekly.

After the MMR Po_2 data were collected, oxygen was again bubbled into the water bath, the flushing pumps were switched on, and the ball valves were opened to reequilibrate respirometer water to near-air saturation conditions. Respirometer water velocity was rapidly increased to 25 cm s^{-1} for 20 min and then slowly increased to 35 cm s^{-1} over 10 min to induce steady swimming. The respirometer was then resealed, and an initial Po_2 sample was removed. Tail-beat and ventilatory frequencies were recorded 15 min into the interval, a final Po_2 sample was collected at 45 min for calculation of a swimming metabolic rate (Mo_2), and the respirometer was unsealed and flushed for 15 min, resulting in a 60-min velocity interval. The velocity was rapidly increased by 5 cm s^{-1} increments, and the protocol was repeated until the fish became exhausted. Exhaustion was defined as three consecutive impingements (60 s

of caudal fin contact with the posterior screen) at the same velocity. The Mo_2 for each interval that resulted in at least a 10-mmHg Po_2 decrease was calculated, with the highest Mo_2 used as the active metabolic rate (AMR). The MSA was then calculated ($\text{AMR} - \text{MMR}$; Fry 1947). This protocol was also used to calculate the U_{crit} (Brett 1964) of green sturgeon using the 60-min velocity intervals.

Liver Collection and Glycogen Assay. Individual fish were rapidly netted and immediately killed by overanesthesia in a buffered MS 222 anesthetic bath (350 mg L^{-1} MS 222, 6 g L^{-1} NaCl, and 0.42 g L^{-1} NaHCO_3) 2 d after the completion of the chronic stress regime ($n = 8$) or after being held in the same tank system for 30 d but not stressed ($n = 8$). After body undulations ceased (ca. 1.5 min), liver tissue was quickly excised, and small pieces were frozen in liquid nitrogen. Samples were then stored at -80°C for measurement of glycogen content.

Liver glycogen levels were analyzed in duplicate by an amyloglucosidase method (Murat and Serfaty 1974; Hung et al. 1989). Approximately 400–500 mg (wet weight) of liver tissue was homogenized in 2 mL of 0.1 M citrate buffer (4°C) containing NaF and centrifuged at 3,000 rpm for 5 min. One set of duplicate aliquots of the clear supernatant was used to determine the total glucose (enzyme digested) in the homogenate, while a second set was used to determine free glucose. The net amount of glucose liberated from liver glycogen was determined by subtracting the total glucose from the free glucose. All samples were measured with a calibrated YSI 2700 biochemical analyzer in triplicate. Standards and a pooled tissue control were used to monitor analyzer and assay accuracy. All procedures were approved by the University of California, Davis, Animal Care and Use Committee (protocol 9368).

Statistical Analysis

Statistical analyses were performed using Sigmapstat 2.03 software (Jandel Scientific, San Rafael, CA). Data from the acute responses to chronic stressors experiment were analyzed either by a one-way ANOVA model followed by Tukey's tests or by a Kruskal-Wallis one-way ANOVA on ranks to determine differences in cortisol levels for a single stressor as well as between the 15-min poststressor responses for all three stressors. Cortisol, glucose, and lactate data from the chronic stress habituation experiment were analyzed by a one-way ANOVA followed by Tukey's tests or *t*-tests to determine differences between prestress and 15-min poststressor plasma concentrations. Differences between the 15-min poststressor plasma concentrations between days (0, 7, 14, and 21) were compared by a one-way ANOVA. Differences between the MMR, MSA, and liver glycogen levels of control and chronically stressed fish were compared using Studentized *t*-tests, and regression analysis was used to compare the initial Po_2 to Mo_2 . Data are reported as means ± 1 SE throughout the study.

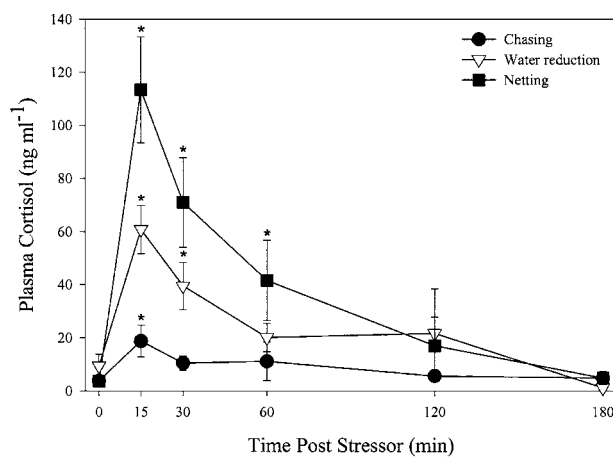


Figure 1. Acute plasma cortisol responses to chronic stressors (chasing, water depth reduction, and net confinement) in cannulated age 0+ green sturgeon. The asterisks denote a significant difference from control. Data are presented as means $\pm$ 1 SE ($n = 6-7$).

Results

Acute Responses to Chronic Stressors

Exposure to the chasing, water depth reduction, and confinement stressors resulted in a graded plasma cortisol response, which reached significantly elevated concentrations of 18.8, 60.8, and 113.4 ng mL⁻¹, respectively (Fig. 1), by 15 min post-stressor. The time needed to return to resting levels appears to be related to the stressor severity, with the times in response to the chasing, water depth reduction, and confinement stressors being 30, 60, and 120 min, respectively (Fig. 1).

Chronic Stress Habituation

Exposure to the chronic stress regime for 21 d did not result in habituation of the green sturgeon's acute stress response, supporting our general hypothesis. Fish remained responsive to a water depth reduction stressor at 0, 7, 14, and 21 d into the chronic stress regime, as determined by significantly elevated ($P < 0.05$, t -test) plasma cortisol concentrations measured at 15 min poststressor (9.4 ± 1.3 , 24.3 ± 4.1 , 20.8 ± 3.6 , and 19.5 ± 2.9 ng mL⁻¹, respectively), when compared with baseline (time 0) levels (2.5 ± 0.3 , 3.6 ± 0.8 , 1.2 ± 0.1 , and 2.4 ± 0.6 ng mL⁻¹, respectively). The cortisol response on day 0 was significantly reduced compared with the responses on days 7, 14, and 21, which were not different ($P < 0.05$, Tukey's test). In addition, the cortisol response measured to the water depth reduction stressor during the "acute response" experiment (60.8 ng mL⁻¹; Fig. 1) was higher than those measured on all days of the "chronic stress" experiment (ca. 20 ng mL⁻¹). We also report significantly higher resting (time 0) glucose levels on days 14 and 21 when compared with day 0 of the chronic stress

regime (Fig. 2) as well as a trend suggesting an acute reduction in lactate at 15 min poststressor throughout the chronic stress regime but only significantly different from the control on day 21 (Fig. 3).

Effects of the Chronic Stress Regime

Growth. There was no significant difference between initial ($1,026.1 \pm 57.8$ g) and final ($1,117.5 \pm 40.8$ g) weights of fish exposed to the 28-d chronic stress regimes, but there was a trend suggesting positive growth during this short period of time. Fish were observed to be actively feeding throughout the chronic stress regime and showed no signs of "poor" health. The weights of control fish were not monitored.

Maintenance Metabolic Rate and Active Metabolic Rate. The 28-d chronic stress regime significantly increased the MMRs of juvenile green sturgeon when compared with control fish, supporting hypothesis 1 (Fig. 4). The AMRs were unaffected by the chronic stress regime (Fig. 4). There was no correlation between initial PO_2 and MO_2 , and solid blocking effects were ignored because the total cross-sectional area of the fish was $<10\%$ of the swimming chamber (Bell and Terhune 1970).

Metabolic Scope for Activity and Critical Swimming Velocity. The MSA was significantly decreased (ca. 25%) in chronically stressed fish when compared with control fish, supporting hypothesis 2 (Fig. 5). Swimming performance, as measured by U_{crit} , was not affected by the chronic stress regime (Table 1). Chronically stressed and control fish (of similar length; Table 1) achieved similar relative sustained swimming speeds, rejecting hypothesis 3.

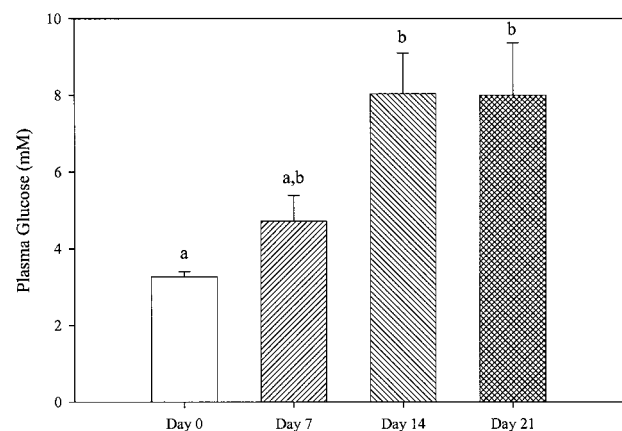


Figure 2. Resting (time 0) plasma glucose levels in age 1+ green sturgeon on days 0, 7, 14, and 21 of a chronic stress regime, which involved exposure to two of three stressors daily for 21 consecutive days. Bars with the same symbol are not significantly different. Data are presented as means $\pm$ 1 SE ($n = 3$).

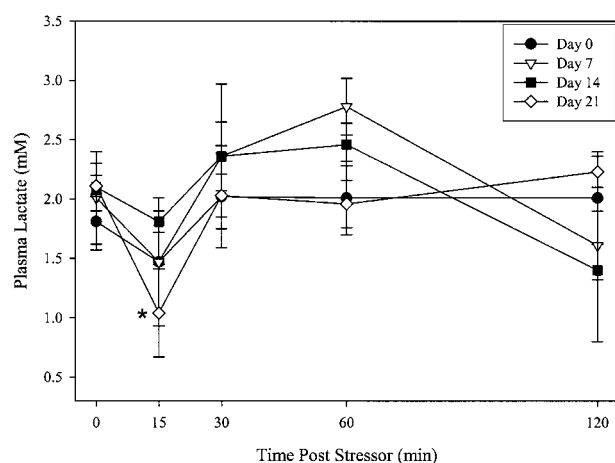


Figure 3. Plasma lactate response to a water reduction stressor in age 1+ green sturgeon on days 0, 7, 14, and 21 of a chronic stress regime, which involved exposure to two of three stressors daily for 21 consecutive days. The asterisk denotes a significant difference from time 0 at day 21. Data are presented as means $\pm$ 1 SE ($n = 3$).

Liver Glycogen. Exposure to the chronic stress regime significantly reduced the green sturgeons' liver glycogen content when compared with control fish (Fig. 6). Mean liver glycogen content decreased by 53% in chronically stressed fish, supporting hypothesis 4, while the free glucose present in the liver tissue before enzymatic breakdown of glycogen was significantly higher in chronically stressed fish (Fig. 6).

Discussion

We investigated the effects of chronic stress on green sturgeon, a chondrosteian fish. The studies described here noted that the hormonal and metabolic effects of chronic stress in green sturgeon differ from the pattern of responses reported in salmonids. Our intent was to quantify the development of the prepathological state (Moberg 1985), which is theorized to be induced by chronic stress and involve increased metabolic demand and vulnerability to additional threats. To study the effects of chronic stress, we first needed to simulate this phenomenon in the laboratory. Past studies have employed cortisol injections, slow-release cortisol implants, miniosmotic pumps, and cortisol-enriched feed to study chronic stress in fish (Barton et al. 1987; Gamperl et al. 1994; Vijayan et al. 1994a; Gregory and Wood 1999). All of these methods resulted in cortisol increases; however, the majority resulted in a persistent increase in cortisol, often to hyperphysiological levels. The effects of these manipulations are not fully understood, but because cortisol participates in a self-regulatory, negative feedback loop (Donaldson 1981; Wendelaar Bonga 1997), they should be considered. A constant elevation of cortisol in nonmammalian vertebrates most likely occurs in only extreme situations; therefore,

we used the modified Shrimpton and Randall (1994) method to simulate chronic stress.

We initially quantified the acute stress response to each of the selected stressors. These stressors were selected to activate the nonspecific stress response with different intensity levels, resulting in a graded cortisol response: the chasing, water depth reduction, and confinement stressors stimulating mild, moderate, and severe response, respectively (Fig. 1). The different intensity levels of stressors were theorized to act as random nonspecific stressors equating to physiologically relevant stressors the fish could experience in the wild. None of these stressors involved air emersion to avoid the development of a respiratory acidosis, which would likely extend the recovery time. Ideally, we would have liked to use stressors that would be more typical in nature, as opposed to the culture environment, but the logistical complications involved in transiently changing pH, temperature, or dissolved O_2 at the available facilities with accuracy proved to be prohibitory. Furthermore, the repetitive use of chemical stressors, which would then have to be cleared from the effluents, would be costly, not to mention the possibility that those compounds might cause unintended insults to the physiology of the fish that would fall outside of the effects of cortisol alone. Clearly, the possibility of a stressor-specific response does exist, but because the intent of this study was to exclusively investigate the cortisol-mediated effect of chronic stress, we feel comfortable with the validity of this model.

After establishing the acute response to the stressors, we determined that green sturgeon did not habituate to our chronic stress protocol. These results differ from the response noted in salmonids, which appear to habituate to chronic stress. For example, rainbow trout exposed to a single daily stressor or to cortisol-enriched feed for 10 wk show habituation of the HPI axis by becoming less responsive to a 30-s air-emersion

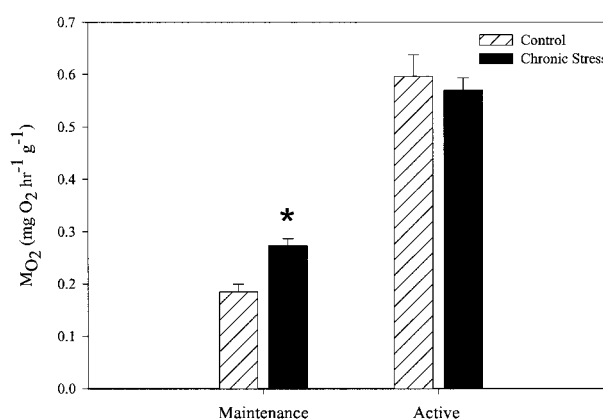


Figure 4. Maintenance and active metabolic rates for age 0+ green sturgeon exposed to a control (unstressed; $n = 11$) or chronic stress ($n = 10$) regime for 28 d. The asterisk denotes a significant difference from the control. Data are presented as means $\pm$ 1 SE.

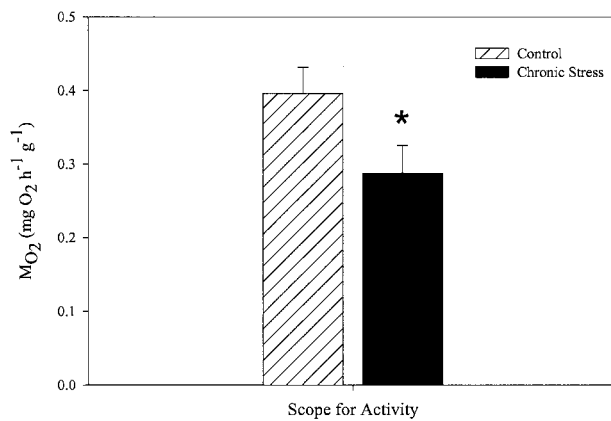


Figure 5. Metabolic scope for activity in age 0+ green sturgeon exposed to a control (unstressed; $n = 11$) or chronic stress ($n = 10$) regime for 28 d. The asterisk denotes a significant difference from the control. Data are presented as means $\pm$ 1 SE.

stressor, despite normal resting cortisol concentrations (Barton et al. 1987). Rainbow trout exposed to a single daily stressor for 6 d have decreased mRNA levels of arginine vasotocin, which is a hypothalamic neuropeptide involved in ACTH stimulation, suggesting habituation at the hypothalamus (Gilchrist et al. 2000). In addition, Sloman et al. (2002) reported a reduced sensitivity of the interrenal cells (in situ) to ACTH and a reduced cortisol synthetic capability in subordinate rainbow trout, which are known to have elevated resting cortisol levels when compared with dominant fish. These results suggest habituation of the HPI axis to chronic stress at the hypothalamus, pituitary, and interrenal tissue levels in salmonids. In contrast, our studies suggest that green sturgeon remained responsive. Interestingly, the cortisol response to the water depth reduction stressor in this experiment was reduced when compared with the response to the identical stressor in the “acute response to chronic stressors” experiment. The use of larger tanks, grouping three fish per tank, and the fact the animals were not cannulated are key differences in the “chronic stress habituation” experiment that could have accounted for the reduced responses. All of these factors would probably reduce the perceived severity of this stressor, yet the fish remained responsive.

The chronic stress regime also affected secondary stress responses in green sturgeon. Resting levels of plasma glucose on days 0 and 7 of the regime were similar (Fig. 2) and consistent with concentrations reported for unstressed green sturgeon (Lankford et al. 2003); Adriatic sturgeon 24 h postcannulation (3.27 mM), as reported by Di Marco et al. (1999); and unstressed hybrid sturgeon (pallid $\times$ shovelnose; 3.28 mM; Barton et al. 2000). However, on days 14 and 21 of the chronic stress regime, resting glucose levels were significantly elevated (Fig. 2), suggesting an increased glucose homeostatic set point in fish exposed to chronic stress. A potential explanation for

this phenomenon could be a prolonged upregulation of gluconeogenesis caused by the repetitive increases of cortisol. However, reduced glucose uptake by peripheral tissues may also account for this change in blood glucose. Additionally, there was a trend for an acute decrease in plasma lactate in response to a water depth reduction stressor (Fig. 3), which is quite different than the response to air emersion or other physically challenging stressors in teleosts and chondrosteans (Mazeaud et al. 1977; Barton et al. 2002). Most reports indicate an increase in lactate levels after exposure to a stressor. Indeed, this response is often accepted as a common secondary measure of stress (Wendelaar Bonga 1997; Barton et al. 2002). Our data suggest that this response is dependent on the nature of the stressor (e.g., requiring a switch from aerobic to anaerobic metabolic processes, consistent with an intense physical challenge or hypoxia exposure caused by air emersion) to elicit the lactate increase. Our results could be explained by increased hepatic clearance rates of lactate and/or use of lactate by gluconeogenesis (Suarez and Mommsen 1987). Despite this novel finding, our chronic stress regime appears to potentiate the post-stressor reduction in lactate at 21 d (Fig. 3) when compared with days 0, 7, and 14. We suggest that this lactate decrease might be caused by a poststressor gluconeogenesis enhancement consistent with the elevated glucose set point. Additional investigations are needed to elucidate these mechanisms more completely, but the limited information available for salmonids does not report similar changes caused by chronic stress (Barton et al. 1987).

Green sturgeon exposed to the chronic stress regime for 28 consecutive days increased their MMR approximately 30%, compared with control fish (Fig. 4), indicating a significant increase in maintenance costs associated with chronic stress. This finding supports our hypothesis as well as the development of the “prepathological state” caused by chronic stress. The MMRs of control fish were similar to that of unstressed Adriatic sturgeon (McKenzie et al. 2001) and to unstressed green sturgeon acclimated to flow-through respirometers (Mayfield and Cech 2004), corroborating the M_{O_2} data from the Brett-type swim-tunnel respirometer used in these experiments. There were no differences between the AMR of chronically stressed or control fish (Fig. 4), unlike the compensatory increase of

Table 1: The critical swimming velocity (U_{crit}) and the total lengths of age 0+ green sturgeon exposed to control (no stress; $n = 11$) or chronic stress ($n = 10$) regimes

	Control	Chronic Stress
U_{crit} (BL s $^{-1}$)	.78 $\pm$.03	.83 $\pm$.02
Total length (cm)	67.6 $\pm$ 1.0	64.8 $\pm$ 1.3

Note. There were no significant differences between the means. Data are presented as means $\pm$ 1 SE. BL = body length.

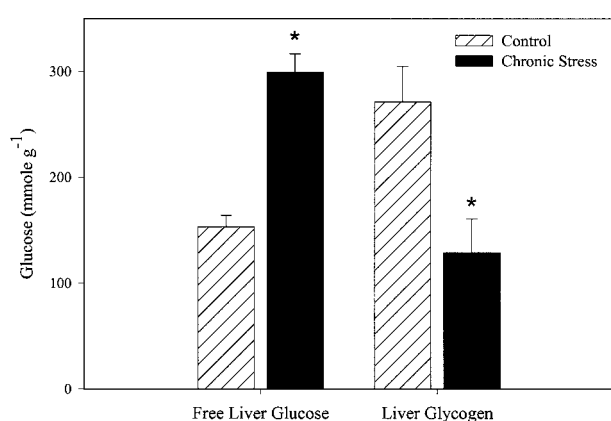


Figure 6. Free hepatic glucose and glucose liberated from hepatic glycogen in age 0+ green sturgeon liver homogenates sampled from either control fish ($n = 8$) or fish subjected to the chronic stress regime ($n = 8$) for 28 d. Asterisks denote a significant difference from the control. Data are presented as means $\pm$ 1 SE.

AMR reported by McKenzie and coworkers (2001) in Adriatic sturgeon exposed to inappropriate salinity.

To estimate the fish's ability to perform work and the effect that chronic stress has on this ability, we calculated the MSA. Exposure to chronic stress significantly decreased the MSA by approximately 25% (Fig. 5), indicating that the increased maintenance cost associated with coping with stress is not compensated for by an increased AMR. Barton and Schreck (1987) investigated the metabolic costs of exposure to an acute stressor immediately before the measurement of MSA in juvenile steelhead. They reported a 25% reduction in MSA associated with the cost of stress, which is remarkably similar to our results. It is worth clarifying that our protocol did not involve an acute stress because all fish were acclimated to the apparatus for at least 16 h, and the control and chronically stressed fish were treated identically. These results suggest that fish exposed to the chronic stress regime would have an attenuated ability to perform work when compared with control fish; the results also provide direct support for the development of the "prepathological state" from a bioenergetic point of view. This reduction in MSA might manifest itself by decreasing reproductive or somatic growth, immunocompetence, or performance-based measures, which all have been investigated as potential effects of chronic stress in fish (Dhabhar and McEwan 1997; Van Weerd and Komen 1998; Gregory and Wood 1999; Haddy and Pankhurst 1999).

To assess changes in a performance variable, we measured the U_{crit} , or maximal sustained aerobic swimming velocity, of control and chronically stressed green sturgeon and found no difference between the two treatments (Table 1), rejecting our hypothesis that chronic stress would reduce U_{crit} . Interestingly, juvenile rainbow trout implanted with cortisol pellets have re-

duced food consumption, growth rates, and food conversion efficiency but have no change in U_{crit} (Gregory and Wood 1999). Adriatic sturgeon exposed to inappropriate salinity had reduced growth rates and increased MMR, but U_{crit} was unchanged (McKenzie et al. 2001), corroborating our results and suggesting that U_{crit} might not be a good measure of chronic stress in fishes. The green sturgeon used in this experiment have a lower U_{crit} compared with teleosts and other species of sturgeon (Adams et al. 1999; McKenzie et al. 2001; Adams and Adams 2003). Sturgeon are generally believed to be less efficient swimmers than teleosts because of the morphology of their heterocercal tail (Lauder 2000; McKenzie et al. 2001); however, our results reveal a lower U_{crit} in green sturgeon compared with other sturgeon, which might be explained by the size of fish investigated and the U_{crit} protocol. Our fish were at least three times the length of fish used in the studies previously cited, and to our knowledge, there are no published results using sturgeon of similar size. In addition, our protocol involved 60-min swimming intervals and did not involve the prodding of fish to remove them from the posterior screen. Although the low U_{crit} values reported here might suggest that green sturgeon are very inefficient swimmers, these data represent the sustained swimming ability of cultured green sturgeon in freshwater and do not represent the "burst" swimming ability or sustained swimming ability of wild green sturgeon in the ocean. The reported U_{crit} for green sturgeon (e.g., 0.83 body lengths s^{-1}) closely approximated those of juvenile scalloped hammerhead sharks (*Sphyrna lewini*; 1.17 body lengths s^{-1} ; Lowe 1996) of approximately the same size.

Finally, we investigated changes in free glucose and glycogen in the livers of chronically stressed and control fish. Chronically stressed fish had approximately 50% more free glucose in the hepatic tissue and vascular space than control fish, while the glycogen content of the liver was reduced by approximately 50% in chronically stressed fish (Fig. 6). These results suggest an increased conversion of glycogen to glucose either for use as a hepatic substrate for enhanced hepatic metabolism, as suggested by Vijayan et al. (1994b), or to be released into the systemic circulation for use by other tissues, both accounting for the increased plasma glucose levels found at days 14 and 21 of our chronic stress regime (Fig. 2). Regardless, it is evident that the source for the increased glucose was at least in part liver glycogen, which was significantly decreased in the stressed fish (Fig. 6). This is somewhat contrary to results in teleosts, which appear to conserve liver glycogen during times of stress by altering hepatocyte responsiveness to glucoregulatory hormones after activation of the stress response (Vijayan et al. 1994a). Barton et al. (1987) also report no change in trout liver glycogen after 10 wk of exposure to a single stressor per day; however, fish fed cortisol-enriched feed for 10 wk did significantly reduce liver glycogen levels in the same experiment, and fish implanted with cortisol pellets had reduced liver glycogen (Vijayan et al. 1994a). Experiments on glycogen changes in

response to acute stressors typically result in significant reductions in liver glycogen (Barton et al. 1986; Vijayan et al. 1994a, 1997). However, these experiments either involved multiple acute stressors applied within 3 h of each other or appear to be short-term changes consistent with glycogenolytic processes caused by catecholamines (Larsson 1973; Randall and Perry 1992). Our results suggest a long-term depletion of liver glycogen in feeding fish that were not stressed for 2 d before liver collection. It is important to point out that our results were probably not caused by the fish going off feed because the fish were observed to be eating on several occasions. The energetic consequences of this glycogen depletion are unknown, but presumably the fish would have 50% less glycogen resources to mobilize during exposure to an acute stressor, reducing the fish's ability to cope with that stressor. It is probable that this situation would be exacerbated if food were not continuously supplied as it was in our experiments.

Taken together, these results suggest that green sturgeon, a "species of special concern" in California, were metabolically compromised by exposure to our chronic stress regime, which we believe is the most accurate method to simulate chronic stress in the laboratory. We argue that the metabolic changes incurred are caused by the inability of the green sturgeon to habituate to the stress regime, as evidenced by significant increases in plasma cortisol throughout the entire stress regime. Because green sturgeon maintain an effective acute stress response, MMR is increased, resulting in a reduction in the MSA. These results suggest an increase in the maintenance requirements because of the additional resources needed to cope with stress, presumably leaving fewer resources for alternative processes such as reproduction, the immune system, growth, and replenishment of energy stores. The decrease in MSA and liver glycogen levels supports the development of the "prepathological state" that is theorized to decrease fitness and survivability of the animal. We feel that these findings, which are not reported in salmonids, might be a factor associated with population declines in this species.

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Literature Cited

- Adams S.R. and G.L. Adams. 2003. Critical swimming speed and behavior of juvenile shovelnose sturgeon and pallid sturgeon. *Trans Am Fish Soc* 132:392–397.
- Adams S.R., J.J. Hoover, and K.J. Killgore. 1999. Swimming endurance of juvenile pallid sturgeon (*Scaphirhynchus albus*). *Copeia* 1999:802–807.
- Barton B.A., H. Bollig, B.L. Hauskins, and C.R. Jansen. 2000. Juvenile pallid (*Scaphirhynchus albus*) and hybrid pallid $\times$ shovelnose (*S. albus* $\times$ *platyrhynchus*) sturgeons exhibit low physiological responses to acute handling and severe confinement. *Comp Biochem Physiol* 126A:125–134.
- Barton B.A., J.D. Morgan, and M.M. Vijayan. 2002. Physiological and condition-related indicators of environmental stress in fish. Pp. 111–148 in S.M. Adams, ed. *Biological Indicators of Aquatic Ecosystem Stress*. American Fisheries Society, Bethesda, MD.
- Barton B.A. and C.B. Schreck. 1987. Metabolic cost of acute physical stress in juvenile steelhead. *Trans Am Fish Soc* 116: 257–263.
- Barton B.A., C.B. Schreck, and L.D. Barton. 1987. Effects of chronic cortisol administration and daily acute stress on growth, physiological conditions, and stress responses in juvenile rainbow trout. *Dis Aquat Org* 2:173–185.
- Barton B.A., C.B. Schreck, and L.A. Sigismondi. 1986. Multiple acute disturbances evoke cumulative physiological stress responses in juvenile chinook salmon. *Trans Am Fish Soc* 115: 245–251.
- Bell W.H. and L.D.B. Terhune. 1970. Water tunnel design for fisheries research. *Fish Res Board Can Tech Rep* 195:1–69.
- Brett J.R. 1964. The respiratory metabolism and swimming performance of young sockeye salmon. *J Fish Res Board Can* 21:1183–1226.
- Campbell P.M., T.G. Pottinger, and J.P. Sumpter. 1994. Preliminary evidence that chronic confinement stress reduces the quality of gametes produced by brown and rainbow trout. *Aquaculture* 120:151–169.
- Carr J.A. 2002. Stress, neuropeptides, and feeding behavior: a comparative perspective. *Integr Comp Biol* 42:582–590.
- Carragher J.F. and J.P. Sumpter. 1990. Corticosteroid physiology in fish. Pp. 487–492 in A. Eppler, C.G. Scanes, and M.H. Stetson, eds. *Progress in Comparative Endocrinology*. Wiley-Liss, New York.
- Cech J.J., Jr. 1990. Respirometry. Pp. 335–362 in C.B. Schreck and P.B. Moyle, eds. *Methods for Fish Biology*. American Fisheries Society, Bethesda, MD.
- Crocker C.E. and J.J. Cech Jr. 1998. Effects of hypercapnia on blood-gas and acid-base status in the white sturgeon (*Acipenser transmontanus*). *J Comp Physiol B* 168:50–60.
- Dhabhar F.S. and B.S. McEwan. 1997. Acute stress enhances while chronic stress suppresses cell-mediated immunity in

- vivo: a potential role for leukocyte trafficking. *Brain Behav Immun* 11:286–306.
- Di Marco P., D.J. McKenzie, A. Mandich, P. Bronzi, E. Cataldi, and S. Cataudella. 1999. Influence of sampling conditions on blood chemistry values of Adriatic sturgeon *Acipenser naccarii* (Bonaparte, 1836). *J Appl Ichthyol* 15:73–77.
- Donaldson E.M. 1981. The pituitary-interrenal axis as an indicator of stress in fish. Pp. 11–40 in A.D. Pickering, ed. *Stress and Fish*. Academic Press, London.
- Fry F.E.J. 1947. Effects of the environment on animal activity. University of Toronto Biology Series 55, publication of the Ontario Fisheries Research Laboratory 68:1–62. University of Toronto Press, Toronto.
- Gamperl A.K., M.M. Vijayan, and R.G. Boutilier. 1994. Experimental control of stress hormone levels in fishes: techniques and applications. *Rev Fish Biol Fish* 4:215–255.
- Gilchrist B.J., D.R. Tipping, L. Hake, A. Levy, and B.I. Baker. 2000. The effects of acute and chronic stresses on vasotocin gene transcripts in the brain of the rainbow trout (*Oncorhynchus mykiss*). *J Neuroendocrinol* 12:795–801.
- Green E.J. and D.E. Carritt. 1967. New tables for oxygen saturation of seawater. *J Mar Res* 25:140–147.
- Greenberg N., J.A. Carr, and C.H. Summers. 2002. Causes and consequences of stress. *Integr Comp Biol* 42:508–516.
- Gregory T.R. and C.M. Wood. 1999. The effects of chronic plasma cortisol elevation on the feeding behaviour, growth, competitive ability, and swimming performance of juvenile rainbow trout. *Physiol Biochem Zool* 73:286–295.
- Haddy J.A. and N.W. Pankhurst. 1999. Stress-induced changes in concentrations of plasma sex steroids in black bream. *J Fish Biol* 55:1304–1316.
- Heath D.D., N.J. Bernier, J.W. Heath, and G.K. Iwama. 1993. Genetic, environmental, and interaction effects on growth and stress response of chinook salmon (*Oncorhynchus tshawytscha*) fry. *Can J Fish Aquat Sci* 50:435–442.
- Hung S.S.O., F.K. Fynn-Aikins, P.B. Lutes, and R. Xu. 1989. Ability of juvenile white sturgeon (*Acipenser transmontanus*) to utilize different carbohydrate sources. *J Nutr* 119:727–733.
- Idler D.R. and B. Truscott. 1972. Corticosteroids in fish. Pp. 126–252 in D.R. Idler, ed. *Steroids in Nonmammalian Vertebrates*. Academic Press, New York.
- Lankford S.E., T.E. Adams, and J.J. Cech Jr. 2003. Time of day and water temperature modify the physiological stress response in green sturgeon, *Acipenser medirostris*. *Comp Biochem Physiol* 135A:291–302.
- Larsson A.L. 1973. Metabolic effects of epinephrine and norepinephrine in the eel, *Anguilla anguilla*. *Gen Comp Endocrinol* 20:155–167.
- Lauder G.V. 2000. Function of the caudal fin during locomotion in fishes: kinematics, flow, visualization, and evolutionary patterns. *Am Zool* 40:101–122.
- Lethimonier C., G. Flouriot, O. Kah, and B. Ducouret. 2002. The glucocorticoid receptor represses the positive autoregulation of the trout estrogen receptor gene by preventing the enhancer effect of a C/EBP β -like protein. *Endocrinology* 143:2961–2974.
- Lowe C. 1996. Kinematics and critical swimming speed of juvenile scalloped hammerhead sharks. *J Exp Biol* 199:2605–2610.
- Maule A.G., R.A. Tripp, S.L. Kaattari, and C.B. Schreck. 1988. Stress alters immune function and disease resistance in chinook salmon (*Oncorhynchus tshawytscha*). *J Endocrinol* 120:135–142.
- Mayfield R.B. and J.J. Cech. 2004. Temperature effects on green sturgeon bioenergetics. *Trans Am Fish Soc* 133:961–970.
- Mazeaud M.M., F. Mazeaud, and E.M. Donaldson. 1977. Primary and secondary effects of stress in fish. *Trans Am Fish Soc* 106:201–212.
- McKenzie D.J., E. Cataldi, P. Romano, S.F. Owen, E.W. Taylor, and P. Bronzi. 2001. Effects of acclimation to brackish water on the growth, respiratory metabolism, and swimming performance of young-of-the-year Adriatic sturgeon (*Acipenser naccarii*). *Can J Fish Aquat Sci* 58:1104–1112.
- Milligan C.L. 1997. The role of cortisol in amino acid mobilization and metabolism following exhaustive exercise in rainbow trout (*Oncorhynchus mykiss* Walbaum). *Fish Physiol Biochem* 16:119–128.
- Moberg G.P. 1985. Biological responses to stress: key to assessment of animal well-being? Pp. 27–49 in G.P. Moberg, ed. *Animal Stress*. American Physiological Society, Bethesda, MD.
- Mommsen T.P., M.M. Vijayan, and T.W. Moon. 1999. Cortisol in teleosts: dynamics, mechanisms of action, and metabolic regulation. *Rev Fish Biol Fish* 9:211–268.
- Murat J.C. and A. Serfaty. 1974. Simple enzymatic determination of polysaccharide (glycogen) content of animal tissues. *Clin Chem* 20:1576–1577.
- Perry S.F., S.G. Reid, and A. Salama. 1996. The effects of repeated physical stress on the β -adrenergic response of the rainbow trout red blood cell. *J Exp Biol* 199:549–562.
- Randall D.J. and S.F. Perry. 1992. Catecholamines. In D.J. Randall and S.F. Perry, eds. *Fish and Physiology: The Cardiovascular System*. Academic Press, New York.
- Reid S. and S. Perry. 1991. The effects and physiological consequences of raised levels of cortisol on rainbow trout (*Oncorhynchus mykiss*) erythrocyte beta-adrenoreceptors. *J Exp Biol* 158:217–240.
- Reid S.G., N.J. Bernier, and S.F. Perry. 1998. The adrenergic stress response in fish: control of catecholamine storage and release. *Comp Biochem Physiol* 120C:1–27.
- Sathiyaa R. and M.M. Vijayan. 2003. Autoregulation of glucocorticoid receptor by cortisol in rainbow trout hepatocytes. *Am J Physiol* 284:C1508–C1515.
- Selye H. 1950. Stress and the general adaptation syndrome. *Br Med J* 1:1383–1392.

- Shrimpton J.M. and D.J. Randall. 1994. Downregulation of corticosteroid receptors in gills of coho salmon due to stress and cortisol treatment. *Am J Physiol* 267:R432–R438.
- Sloman K.A., C.J. Montpetit, and K.M. Gilmour. 2002. Modulation of catecholamine release and cortisol secretion by social interactions in the rainbow trout (*Onchorhynchus mykiss*). *Gen Comp Endocrinol* 127:136–146.
- Suarez R.K. and T.P. Mommsen. 1987. Gluconeogenesis in teleost fishes. *Can J Zool* 65:1869–1882.
- Sumpter J.P. 1997. The endocrinology of stress. Pp. 95–118 in G.K. Iwama, A.D. Pickering, J.P. Sumpter, and C.B. Schreck, eds. *Fish Stress and Health in Aquaculture*. Society for Experimental Biology Seminar Series 62. Cambridge University Press, Cambridge.
- Van Eenennaam J.P., M.A.H. Webb, X. Deng, S.I. Doroshov, R.B. Mayfield, J.J. Cech, D.C. Hillemeier, and T.E. Willson. 2001. Artificial spawning and larval rearing of Klamath River green sturgeon. *Trans Am Fish Soc* 130:159–165.
- Van Weerd J.H. and J. Komen. 1998. The effects of chronic stress on growth in fish: a critical appraisal. *Comp Biochem Physiol* 120A:107–112.
- Vijayan M.M., C. Pereira, G.E. Grau, and G.K. Iwama. 1997. Metabolic responses associated with confinement stress in tilapia: the role of cortisol. *Comp Biochem Physiol* 116C: 89–95.
- Vijayan M.M., C. Pereira, and T.W. Moon. 1994a. Hormonal stimulation of hepatocyte metabolism in rainbow trout following an acute handling stress. *Comp Biochem Physiol* 108C:321–329.
- Vijayan M.M., S. Raptis, and R. Sathiyaa. 2003. Cortisol treatment affects glucocorticoid receptor and glucocorticoid-responsive genes in the liver of rainbow trout. *Gen Comp Endocrinol* 132:256–263.
- Vijayan M.M., P.K. Reddy, J.F. Leatherland, and T.W. Moon. 1994b. The effects of cortisol on hepatocyte metabolism in rainbow trout: a study using the steroid analogue RU486. *Gen Comp Endocrinol* 96:75–84.
- Wedemeyer G.A., B.A. Barton, and D.J. McLeay. 1990. Stress and acclimation. Pp. 451–489 in C.B. Schreck and P.B. Moyle, eds. *Methods for Fish Biology*. American Fisheries Society, Bethesda, MD.
- Wendelaar Bonga S.E. 1997. The stress response in fish. *Physiol Rev* 77:592–616.